

Mohamed Kamel

Richmond, VA 23219, United States

 (+1) 8043967089  kamelms@vcu.edu  Mosalah022  mohamed-kamel

Education

Virginia Commonwealth University (VCU)

PH.D. IN COMPUTER SCIENCE

- GPA: 4.0/4.0

Richmond, VA, United States

2024 – Present

Nile University (NU)

M.Sc. IN INFORMATICS

- GPA: 3.97/4.0

Cairo, Egypt

2022 – 2024

Ahram Canadian University (ACU)

B.Sc. IN COMPUTER SCIENCE AND INFORMATION TECHNOLOGY

- GPA: 3.9/4.0, Second of the class

Cairo, Egypt

2018 – 2022

Skills

Programming

Expert: Java, Python, C++, JavaScript, PHP, HTML, CSS | Familiar: Dart, Kotlin

Backend Dev

FastAPI, Docker, REST APIs

ML & DL

Reinforcement Learning, Self-Supervised Learning, Transfer Learning

Vision

Computer Vision, ViT, BEiT, Semantic Segmentation, Semantic Communication, U-Net, GANs, Diffusion Models, DINOv2

LLMs

Prompt Engineering, LLM Poisoning, LoRA, TinyLLaMA, GPT, Claude, NER

Databases

Expert: MySQL, MongoDB | Proficient: Firebase

Frameworks

Expert: JavaFX, Swing, Bootstrap | Proficient: JSON, Flutter | Familiar: Kafka, Microservices

Languages

Arabic: Native | English: C1 | German: A1

Publications

BeatFlow-ECG: Rectified Flow for ECG Reconstruction from Indirect Wearable Signals

CONFERENCE PUBLICATION

NeurIPS

Under Review, 2026

ProSeD: Source-Side Semantic Decoys Against Unauthorized Data Harvesting

JOURNAL PUBLICATION

Knowledge-Based Systems (KBS)

Under Review, 2026

CondFlow-ECG: Conditional Rectified Flow for Morphology-Aware ECG Denoising

CONFERENCE PUBLICATION

International Conference on
Machine Learning and Applications
(ICMLA)

Accepted, 2026

Development of an AI-Enabled Remote Loneliness Assessment System for Older Adults Using Behavioral and Speech Data from IoT Sensors and Smart Speakers

JOURNAL PUBLICATION

Assessment (SAGE), Special Issue on
AI in Psychological Assessment

Accepted, 2026

Multimodal Activity Episode Inference from Smart-Home Signals for Behavioral Health Monitoring in Older Adults

CONFERENCE PUBLICATION

International Workshop on Machine
Learning for Signal Processing
(MLSP)

Under Review, 2026

URGE: Urgency-Aware Energy-Efficient UAV Mobility via Hybrid Deep Reinforcement Learning

CONFERENCE PUBLICATION

International Conference on Smart
Mobility (SM) [\[View Paper\]](#)

2026

Attention-Driven Reinforcement Learning to Optimize Packet Retransmission in Semantic Communication

CONFERENCE PUBLICATION

IEEE GLOBECOM [\[View Paper\]](#)

2025

SkinFormer: Robust Vision Transformer for Automatic Skin Disease Identification

CONFERENCE PUBLICATION

JAC-ECC [\[View Paper\]](#)

2022

Academic Service

IET Communications journal

REVIEWER

2026

International Conference on Machine Learning and Applications (ICMLA)

REVIEWER

2026

Assessment journal

REVIEWER

2026

International Conference on Smart Mobility (SM)

REVIEWER

2026

Research Talks

From Wearable Signals to Clinical ECG: Toward Continuous Cardiac Monitoring

INVITED SPEAKER

Department of Computer Science
Seminar, Virginia Commonwealth
University

March 20, 2026

Honors, Awards & Certifications

Badge issued by Virginia Commonwealth University (VCU)

LEADERSHIP FOUNDATIONS FOR GRADUATE STUDENTS

[Credly](#)

Oct 2025

Credential from IBM Z Day on AI and Data

IBM Z DAY 2025 SE – AI & DATA

[IBM](#)

Apr 2025

Credential from MongoDB

MONGODB CRUD OPERATIONS IN PYTHON

[MongoDB University](#)

2025

Semi-Finalist, Dell Technologies Envision the Future

GRADUATION PROJECT COMPETITION (MIDDLE EAST, RUSSIA, AFRICA, TURKEY)

[Online](#)

2025

Certificate, Software Design and Architecture Specialization University of Alberta

COVERED: OBJECT-ORIENTED DESIGN, DESIGN PATTERNS, SOFTWARE ARCHITECTURE, SERVICE-ORIENTED ARCHITECTURE

[Online](#)

2021

First Place, ACU Problem Solving Competition

HACKEREARTH COLLEGE CHALLENGE

[Leaderboard](#)

2020

Academic Excellence Award

AWARDED MULTIPLE TIMES FOR TOP ACADEMIC PERFORMANCE

[Ahram Canadian University](#)

2019 – 2022

Experience

Virginia Commonwealth University (VCU) RESEARCH ASSISTANT - Contributing to the NIH-funded <i>HomePAL</i> project on AI-enabled ambient health monitoring (NIH R01AG072611, Project Link). - Responsible for sensor setup, data acquisition scripts, signal analysis, and AI-based behavioral anomaly detection - Contributing to the SARE-funded food waste tracking and diversion project (SARE FLW24-004, Project Link).	<i>United States</i> Aug. 2025 – Present
VCU + Bank of America – Special Olympics SOFTWARE ENGINEER INTERN (PART-TIME) - lead the backend development of a software application supporting Special Olympics. - Involved in software design, API development, testing, and integration with frontend components. - Tech Used: Python, FastAPI, MongoDB, Azure, Figma, React, TypeScript, HTML, CSS, JavaScript, Git	<i>United States</i> May 2025 – Aug 2025
Virginia Commonwealth University (VCU) GRADUATE ASSISTANT (FULL-TIME) - Held weekly office hours, and provided academic support for foundational CS courses. - Graded assignments and assisted in curriculum delivery for undergraduate students.	<i>United States</i> Jan. 2024 – May 2025
Nile University (NU) TEACHING ASSISTANT (FULL-TIME) - Assisted in teaching AI, CS major, and programming courses. - Provided lab support and one-on-one mentoring for students.	<i>Egypt</i> Feb. 2023 – Jan. 2024
Ahram Canadian University (ACU) TEACHING ASSISTANT (FULL-TIME) - Taught AI-related and core programming subjects. - Supported student learning and practical lab implementation.	<i>Egypt</i> Oct. 2022 – Feb. 2023
LÀCRIMA Dairy Industrial Ltd SOFTWARE DEVELOPER (CONTRACT) - Developed an internal desktop system for electronic file archiving and communication (Demo Video). - Designed customer-facing and employee mobile applications to improve service interaction.(Android iOS) - Tech Used: Java, JavaFX, AWS, MySQL, Flutter, Firebase, Kotlin	<i>Remote - Jordan</i> Jan. 2022 – Jul. 2023
Order GO FULL STACK DEVELOPER (FULL-TIME) - Built the company's public website and improved digital presence. - Developed the ROS App for streamlining kitchen order management. - Tech Used: Java, JavaFX, Kotlin, HTML, CSS, JavaScript, Bootstrap, PHP, Firebase, Git	<i>Remote - United Kingdom</i> Apr. 2021 – Oct. 2021
Upwork TOP-RATED FREELANCER - Completed 65+ freelance software projects with a 98% Job Success Score. - Delivered AI, web, mobile, and desktop solutions for international clients.	<i>Freelance</i> Apr. 2019 – Oct. 2023

Volunteering Activities

Ahram Canadian University (ACU) VOLUNTEER UNDERGRADUATE TEACHING ASSISTANT - Supported courses: Linear Algebra, Fundamentals of Programming I, Event Driven, and Data Structures. - Uploaded educational videos to YouTube; one video received nearly 50,000 views with excellent feedback (Video Link).	<i>Egypt</i> 2018 – 2022
EXCEPTION Student Committee MEMBER AND ACADEMIC INSTRUCTOR - First academic and technical-only content committee in ACU CS. - Conducted workshops in multiple areas including CS fundamentals, programming, Java (OOP), and Web development.	<i>Egypt</i> 2018 – 2022
ACU CPC Community (ACM) MENTOR Mentored peers in programming competitions and problem-solving practices.	<i>Egypt</i> 2020 – 2022
ACU CS Student Union SCIENTIFIC COMMITTEE MEMBER Supported the academic initiatives and contributed to scientific activities organized by the student union.	<i>Egypt</i> 2020 – 2021

- Participated in Google Developer Student Club activities, workshops, and peer-learning sessions.

Selected Projects

ENERGY-EFFICIENT UAV DEPLOYMENT VIA REINFORCEMENT LEARNING FOR DISASTER RESPONSE

2024

- Developed a UAV deployment framework prioritizing disaster tasks with Deep Q-Networks (DQN).
- Integrated collaborative drone behavior to maximize coverage and minimize energy use.
- Enhanced efficiency of UAV interventions through dynamic prioritization.
- Technologies: Reinforcement Learning, DQN, UAVs, Swarm Intelligence, Python.

NAVIGATING THE RETINAL MAZE: IR IMAGING FOR RETINAL EXPLORATION

2024

- Developed CrackRAVIR, a deep learning model for segmentation of retinal arteries and veins using IR imaging.
- Utilized Residual Attention UNET to improve segmentation accuracy and vascular width estimation.
- Improved glaucoma and AMD detection through detailed vasculature analysis.
- Technologies: Deep Learning, Residual Attention UNET, Semantic Segmentation, Python, IR Imaging.

GRAPH NEURAL NETWORKS FOR MOVIE RECOMMENDATION SYSTEMS

2024

- Developed a GraphSAGE-based recommendation system for movies using bipartite graphs of users/movies.
- Addressed cold start and scalability issues, compared with collaborative filtering and LightGCN.
- Technologies: Python, GraphSAGE, PyTorch Geometric, Collaborative Filtering, LightGCN.

EVIDENCE DETECTION IN CLOUD FORENSICS

2023

- Designed a system to detect evidence of cyber threats and anomalies in cloud environments.
- Utilized Gradient Boosting Machines, Sequential NN, and RNN to improve forensic accuracy and speed.
- Technologies: Python, GBM, RNN, Enhanced MLP.

AI DISEASE DIAGNOSTIC SYSTEM USING DEEP LEARNING AND IMAGE PROCESSING

2022

- Developed an application for early diagnosis of diseases using 50,000+ images from Fitzpatrick17k, SD198, and ISIC 2019 datasets.
- Achieved 85% accuracy using LAMB optimizer and advanced image augmentation techniques.
- Built with clean architecture and MVI framework.
- Technologies: Machine Learning, Deep Learning, Python, Kotlin, Vision Transformers, Neural Networks, MVI.
- [Demo Preview](#) | [Paper](#)

SELF-DRIVING CAR WITH OBJECT DETECTION (CNN, RL, YOLOv3)

2022

- Developed a self-driving car model capable of detecting traffic signals and adjusting speed dynamically.
- Technologies: Computer Vision, Carla Simulator, CNNs, Reinforcement Learning, YOLOv3, Python.

EXTENDING FASTPITCH WITH EMOTIONAL AWARENESS

2021

- Enhanced Fastpitch for emotional speech output using a modified Conformer architecture.
- Trained with emotional labels and perceptual loss for expressive speech synthesis.
- Technologies: Python, Conformer, Fastpitch, DeepVoice3, Mel Spectrogram.

EARLY-STAGE DIABETES RISK PREDICTION CLASSIFIER

2021

- Developed a binary classifier for diabetes risk prediction using tabular data.
- Technologies: Python, FastAI, Machine Learning.
- [GitHub Repo](#)

WHEAT SEEDS CLASSIFICATION

2021

- Classified wheat seeds using multiple machine learning algorithms, including Naïve Bayes, Random Forest, SVM, Decision Tree, and K-Nearest Neighbors.
- Compared performance across models to determine the most accurate classifier.
- Technologies: Python, Naïve Bayes, Random Forest, SVM, Decision Tree, KNN.
- [GitHub Repo](#)

ACCOUNT MANAGEMENT SYSTEM

2020

- Automated accounting tasks, including employee hour tracking, payroll generation, and bonus/deduction calculations.
- Technologies: Java, JavaFX, MySQL.

HOTEL RESERVATION SYSTEM

2020

- Developed a Java desktop application for hotel receptionists to manage room reservations, services, and guest details, including payment tracking.
- Simplified hotel operations by automating room management and customer billing.
- Technologies: Java, JavaFX, MySQL.

TAZKARTI FOOTBALL MATCH BOOKING SYSTEM

2019

- Developed a user-friendly system to register, select matches, and book tickets online.
- Technologies: Java, JavaFX, MySQL.

OTHER PROJECTS (SELECTION)

- E-Learning School Website – HTML, CSS, JS, PHP, SQL. [Demo](#)
- Hospital Management System – Java, JavaFX, MySQL. [Demo](#)
- Inventory Management System – Java, JavaFX. [Demo](#)
- Learn Tech Website – HTML, CSS, JS, Bootstrap. [Preview](#)
- Restaurant Website – HTML, CSS, JS. [Demo](#)
- Grow Social App – Java, JavaFX, MySQL. [Demo](#)
- Games: Scotland Yard ([Demo](#)) | Snake 2D ([Demo](#)) | Memory ([Demo](#)) | Kids Games (Java, JavaFX, MySQL) | Hangman (HTML, CSS, JavaScript)

Key Skills & Competencies

- Fast Learner of New Technologies and Programming Languages
- Excellent Communication and Team Collaboration
- Strong Analytical and Problem-Solving Skills
- Performs Well Under Pressure and Shares Knowledge Effectively

Additional Information

Hobbies	Reading research papers on AI and Computer Vision, participating in coding challenges (e.g., Hackerrank, LeetCode), contributing to open-source projects, attending tech meetups and AI webinars, gym, football, and classical music
Social Activities	Helping people and sharing knowledge

References

Available upon request.